Title Page

Protocol Title: The Norwegian WHO Labour Care Trial, a stepped wedge cluster randomised trial

Protocol Number: NA

Compound Number: NA

Short Title: NORWELCG

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Regulatory Agency Identifier Number(s):

Registry	**ID**
ClinicalTrials.gov	NCT05791630

SAP Version: 1.0, 14 October 2026

Signature Page

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Version History

This statistical analysis plan (SAP) for NORWELCG is based on protocol version 1.06, dated 9 May 2024.

SAP Version	Date	Change	Rationale
1.0	14.10.2026	Not applicable	Original version

List of Abbreviations

Abbreviation	Definition
BD / BE	Base deficit / base excess
BMI	Body mass index
CC	Complete case
CEQ2	Childbirth Experience Questionnaire, version 2
CTG	Cardiotocography
eCRF	Electronic case report form
FAS	Full analysis set
HC3	Heteroscedasticity-consistent standard errors, type 3
ICC	Intraclass correlation coefficient
ICH	International Council for Harmonisation
IM / IV	Intramuscular / intravenous
LCG	Labour Care Guide
MAR	Missing at random
NORBIRTH	Norwegian Network for Obstetric Research
OASIS	Obstetric anal sphincter injury
PAS	Primary analysis set
SAP	Statistical analysis plan
STAN	ST analysis of the fetal electrocardiogram
TTE	Time to event
VDS	Vaginal delivery set

Abbreviation	Definition
WHO	World Health Organization

1. Introduction

The Norwegian WHO Labour Care Trial (NORWELCG) is a pivotal multicenter, stepped wedge cluster randomised trial aimed at evaluating the effect of the Labour Care Guide (LCG) compared to the traditional WHO partograph in Norwegian hospitals. This study seeks to assess the effects of LCG on intrapartum interventions, maternal and neonatal outcomes, and healthcare provider and patient experiences.

The trial adopts the principles established in the ICH E9 Statistical Principles for Clinical Trials and its addendum E9 (R1) focusing on estimands and sensitivity analyses. This SAP will elucidate the statistical methodologies intended for estimating treatment effects, aligning the analysis with the outlined estimands, thus ensuring a robust interpretation of trial objectives.

Protocol Changes: The SAP reflects the original design of the trial as detailed in protocol version 1.06 (dated 2024.05.09). There are currently no changes to protocol-planned analyses; future amendments will be documented in Section 10, underscoring transparent methodological evolutions responsive to trial conduct.

Exclusions and Supplementary Documentation: While primary and secondary analysis specifications are contained herein, other analyses and descriptive statistics such as baseline summaries, exploratory endpoints, and detailed demographic analyses are not necessarily included. Some secondary analyses are considered for secondary publications and are further specified in topic-specific statistical analysis plans.

1.1 Objectives, Endpoints, and Estimands

Objectives and Endpoints

Further objectives and endpoints as described in the protocol but to be specified in other analyses plans are provided in Section 12.1

Table 1: Objectives and endpoints

Objectives	**Endpoints**
Primary	
To investigate the effect of LCG in women with a planned vaginal delivery on intrapartum caesarean sections	Occurrence of intrapartum caesarean sections
Secondary	
To investigate the effect of LCG in women with a planned vaginal delivery	Occurrence of Spontaneous vaginal delivery
	Occurrence of Instrumental vaginal delivery

Objectives	**Endpoints**
	Occurrence of amniotomy during labour*
	Use of oxytocin during labour*
	Use of epidural analgesia during labour*
	Occurrence of intermittent fetal monitoring*
	Occurrence of continuous fetal monitoring*
	Occurrence of OASIS
	Occurrence of episiotomy
	Occurrence of postpartum haemorrhage > 1000 ml*
	Time from 4 cm cervical dilation to vaginal delivery
To investigate the effect of LCG on the neonate	Arterial umbilical cord pH < 7.00*
	Apgar score < 7 at 5 minutes*
	Admittance to neonatal intensive care unit within 2 hours after birth*
Exploratory	
To describe and explore the effect of LCG in women with a planned vaginal delivery on other clinical aspects of intrapartum caesarean sections	Cervix dilation at time of caesarean section#
	Level of the presenting part at caesarean section#
	Fetal presentation at caesarean section#
	Fetal position at caesarean section#

Objectives	Endpoints
	Main reason for caesarean section#
	Type of continuous fetal monitoring*
	Occurrence of prolonged first or second stage of labour according to the different definitions#
	Occurrence of shared decision making assessed in selected situations and interventions and documented by the midwife#

* Detailed in other statistical analysis plans

Only descriptive, detailed in other statistical analysis plans

1.2 Estimand for Primary Objective

Primary Objective: To investigate the effect of LCG in women with a planned vaginal delivery on intrapartum caesarean sections

Estimand: The primary estimand in this trial is the absolute difference in proportion of intrapartum caesarean sections between women with a planned vaginal delivery in need of a partograph following the LCG compared to the WHO partograph

Table 2: Estimand to analysis table

Estimand	Analysis
Target population	Analysis set
Adult pregnant women with a planned vaginal delivery	All women ≥ 18 years admitted for a planned vaginal delivery with a live fetus at admission at any of the participating hospitals in the period 01.04.2025-26.05.2026. Women who do not wish to participate in the trial will be excluded from the analysis set. The participating hospitals are: Østfold Hospital Trust, University Hospital in Northern Norway, Vestre Viken Hospital Trust: Drammen, Stavanger University Hospital, Sørlandet Hospital: Kristiansand, Oslo University Hospital: Ullevål, Oslo University Hospital: Rikshospitalet, St. Olav University

Estimand	**Analysis**
	Hospital, Akershus University Hospital and Haukeland University hospital. The primary analysis set (PAS, Section 3) excludes women delivering during their hospital's transition period (Figure 1).
Treatment	
The LCG compared with the WHO partograph. The 5-week transition (teaching) period in each hospital belongs to neither condition.	Treatment assignment is determined by the eCRF, which changed at each participating hospital's crossover date to the LCG (Section 3). The crossover date was randomised as detailed in Section 1.4. Women delivering during their hospital's transition period are excluded from the primary analysis set. Including them would bias the effect towards no difference, which is anti-conservative for the non-inferiority assessment. An analysis including the transition periods as control is described in Section 4.1.5.
Variable	Outcome measure
Delivery by caesarean section Dichotomous (yes/no)	Intrapartum caesarean sections as recorded in the hospital's medical records
Handling of intercurrent events	Handling of missing data
Some hospitals may be influenced to use whole of parts of LCG even before transition from WHO Partograph to LCG. Strategy: Treatment policy approach—treat according to assigned strategy regardless of crossover timing	None found. At the blinded data review of the final data export (1 October 2026; 36,322 women), mode of delivery was recorded for all women, and cervical dilatation for every intrapartum caesarean section. Should the locked database contain missing values, they will be ignored (complete-case analysis).
The WHO partograph defines active labour from 4 cm cervical dilatation and the LCG from 5 cm, so which women are "in need of a partograph", and which caesarean sections are	

Estimand	Analysis
intrapartum, depends on the tool. To define the primary endpoint identically in the two periods, all caesarean sections from 4 cm cervical dilatation are counted, whatever the period. Women delivered by caesarean section before reaching 4 cm were not included in the trial. A caesarean section recorded at 3 cm, after the cervix had retracted (5 women), is counted with those at 4 cm.	
Population-level summary measure	Analysis approach
The primary summary measure for treatment effect will be the absolute difference in proportion of the target population with an intrapartum caesarean section. A supplementary summary measure will be the relative difference.	The risk difference is computed based on a generalised linear mixed logistic model with a random term for cluster and fixed effect for treatment and time interval. The absolute risk difference (primary) and the relative difference (supplementary) are calculated by G-computation as population-averaged (marginal) adjusted estimates, with standard errors estimated by the delta method.

1.3 Estimands for Secondary Objectives

All estimands for secondary objectives have the following attributes:

- The target population is adult pregnant women with a planned vaginal delivery in need of a partograph
- The investigational intervention is using the LCG, as compared to the WHO partograph during labour. Treatment is defined by the eCRF, which each hospital changed at its crossover date in the randomised schedule (Figure 1, Section 3). Women included during the 5-week transition (teaching) period in each hospital will be excluded from the primary analysis set.
- Any intercurrent events, such as not complying to assigned labour guideline will be handled by the treatment policy strategy, i.e. treat according to assigned strategy regardless of compliance
- As for the primary endpoint, all women in the primary analysis set are included, including those delivered by caesarean section at 4 cm, unless specified otherwise for an endpoint.
- Amniotomy, oxytocin and epidural analgesia are counted whatever the cervical dilatation at which they were performed or started (Section 5.1.1.1).

- For endpoints that exist only after a vaginal birth (episiotomy, OASIS), a caesarean section is an intercurrent event that precludes the outcome. It is handled by a principal stratum strategy, i.e. the stratum of women who would deliver vaginally with either labour-monitoring tool, estimated in the vaginal delivery set (Section 3). The causal interpretation of this relies on the LCG not changing which women deliver vaginally. A supplementary analysis uses a composite strategy, counting a caesarean section as no event.
- In time from 4 cm cervical dilatation to vaginal delivery, a caesarean section is handled by a composite strategy where it is included as no vaginal delivery during the first 36 hours after 4 cm regardless of when the caesarean section happened.
- Population-level summary measures will be
    - Absolute and relative difference in proportion for dichotomous endpoints
    - Difference in restricted mean time without vaginal delivery during the first 36 hours after 4 cm for time from 4 cm cervical dilatation to vaginal delivery

Further details are given in Section 5.

1.4 Study Design

From the protocol section 4: The NORWELCG study is conducted as a stepped wedge cluster randomised trial. The choice of cluster randomisation secures a standard guideline for the entire department/unit as there would be a risk of contamination if randomising on an individual level, given that the same midwife/obstetrician could care for women with different guidelines at the same time. The choice of a stepped wedge cluster randomised design is based on the possible risk of the Hawthorne effect in a standard cluster randomised trial, experienced when conducting the multicentre cluster randomised trial: The Labour Progression Study in Norway 2014-2018 (17). By allowing all units to use both the WHO partograph and the LCG in steps, the risk of bias represented by contamination and the Hawthorne effect will be reduced to a minimum. A non-inferiority approach has been chosen to evaluate the effect of the LCG. Labour outcomes are overall good in Norway and a possible national enrolment of the LCG should be based on results similar or better than we already have. The choice of intrapartum caesarean section rate as main outcome is due to its clinical significance and the potential to merge results with other ongoing LCG studies on research priority questions stated by the WHO.

Randomisation

Prior to trial commencement, the clusters were randomly assigned to one of the sequences (S1-S10) for time of transition from WHO partograph to the LCG (Figure 1), using a computer-generated randomisation system. To obtain an equal distribution of small and larger hospitals during the study period, randomisation was stratified by hospital size. To obtain an equal distribution of intrapartum caesarean section rates between units, randomisation was also stratified by the rate of intrapartum caesarean sections prior to onset of the trial. The allocation sequence was masked for the units until the unit is in turn for transition to the LCG, to avoid contamination.

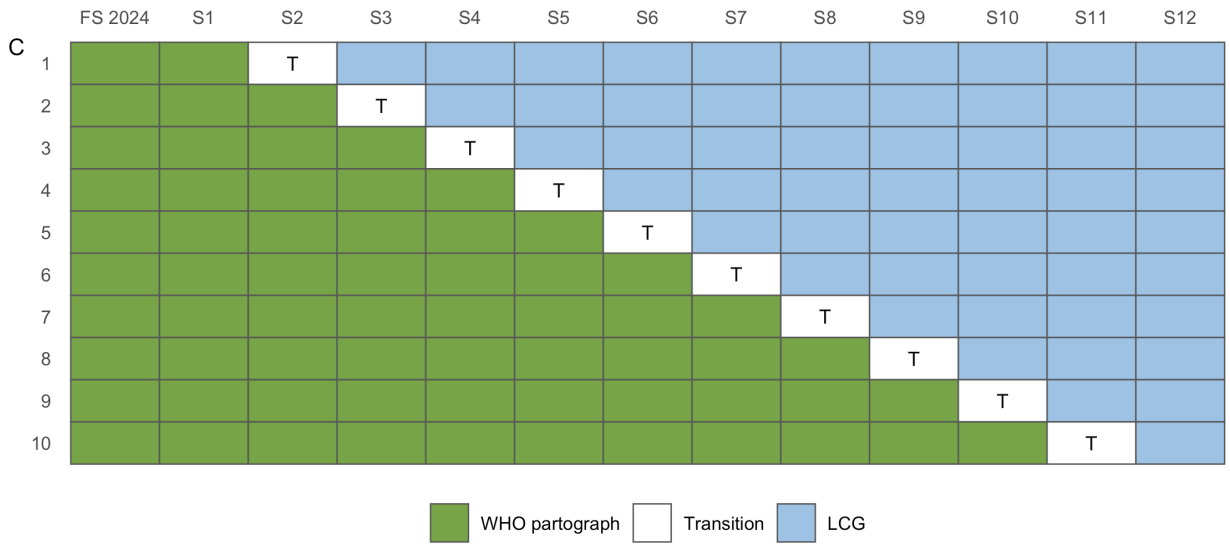


Figure 1: **Overview of the study steps and transition periods.** C = Cluster, FS = feasibility study of the electronic version of the LCG. S = steps of 5 weeks (S1-S12 = 60 weeks). Green colour denotes WHO partograph period, white colour denotes transition period for and teaching period, and blue colour denotes LCG period

2. General Considerations

As stated in the protocol, this SAP has been written and approved prior to commencement of the analysis and without knowledge of allocation. Amendments to plan can be done until day of data base lock. All statistical analysis code has been written blinded to the participant's treatment allocation to assess the appropriateness and validity of the prespecified statistical analyses. Any revisions until the final version 1.0 will be based on blinded review of the data. Revisions after unblinding will be clearly marked, and versioning control both on the code and SAP will show all analysis decisions in a transparent way.

2.1 Statistical Hypotheses and Conclusion Criteria

The primary hypothesis is one of non-inferiority: the WHO partograph is the standard of care and the LCG the new approach, and a lower caesarean section rate with the LCG would be a favourable outcome. With the difference defined as the proportion of intrapartum caesarean sections with the LCG minus that with the WHO partograph, among women with a planned vaginal delivery (primary estimand, Section 1.2), the null hypothesis is

- H_0: the difference is greater than or equal to 1.5 percentage points (the LCG is inferior),

and the alternative hypothesis is

- H_1: the difference is less than 1.5 percentage points (the LCG is non-inferior).

The non-inferiority margin is set at 1.5 percentage points. The null hypothesis is rejected, and the LCG concluded non-inferior to the WHO partograph, if the upper limit of the two-sided 95% confidence interval for the difference lies below 1.5 percentage points. This corresponds to a one-sided test at the 2.5% significance level.

The rationale for choosing 1.5 percentage points as non-inferiority margin was based on a discussion within the trial steering committee consisting of experienced clinicians, midwives and a statistician.

If non-inferiority is shown, superiority of the LCG is tested next, in a fixed sequence and without adjustment of the significance level. The LCG is concluded superior with respect to intrapartum caesarean sections if the upper limit of the same confidence interval also lies below 0.

2.2 Multiplicity Adjustment

The non-inferiority and superiority tests of the primary estimand form a fixed sequence. Superiority is tested only if non-inferiority has been shown, which keeps the overall one-sided type I error at 2.5% without adjustment. There will be no adjustments for multiplicity for the secondary endpoints.

2.3 Impact of Intercurrent Events Strategies

The treatment policy strategy implies that we are looking at the effect of intending to use the interventional treatment (using the LCG guideline) regardless of actual implementation or compliance to the protocol. For the primary endpoint, all caesarean sections from 4 cm cervical dilatation are counted in both periods (Section 1.2). For episiotomy and OASIS, which exist only after a vaginal birth, a principal stratum strategy is used (Section 1.3).

2.4 Handling of Missing Data

Generally, there will be three methods to deal with missing data:

1) Using multiple imputation where the imputation is based on a model with baseline information such as women's age, gestational age, hospital, treatment and time interval as fixed effects
2) Imputing with worst outcome
3) Ignoring missing data

The last two approaches will only be used in cases where there are very few missing data. The precise handling of missing data will be detailed by endpoint/estimand in section 4 and 5. The handling of missing data will be revised before database lock and final signed version of this SAP. This revision will be based on a blinded look at the data, and any changes to the handling will not be based on any unblinded inference.

Missing values of covariates used for adjustment are handled by the missing-indicator method (Section 4.1.3).

2.5 Use of AI Tools

The statistician used Claude Code (Anthropic), an AI coding assistant based on large language models, in the development of this SAP and of the analysis code. It was used to draft and revise SAP text, to check consistency between the SAP, the analysis code and the mock-up report, to write and test the analysis code, and to verify cited references against their full texts. All analytical decisions were made by the statistician and the investigators; every AI-generated text and code change was reviewed by the statistician, who takes full responsibility for the content. Claude Code worked only in a development environment with an anonymised copy of the data (study period, hospital, timings and outcomes permuted or shifted, record keys replaced); it had no access to identifiable data or to the true allocation, and the analyses of the real data are run in a secure environment without access to AI tools. Decisions and their rationale are documented in a version-controlled review log kept with the analysis code.

3. Analysis Sets

Three analysis sets are defined. Whether a woman belongs to the control, transition or LCG period is determined by the eCRF, which is directly linked to the change of guideline. The LCG eCRF was taken into use at 08:00 on the hospital's crossover date in the randomised schedule (Figure 1): women who entered active labour after that time and were documented on the LCG form belong to the LCG period, and women already in labour and documented on the WHO partograph form belong to the preceding period, even if they gave birth later. The WHO partograph form was used in both the control and the transition period; the two are separated by the hospital's transition date in the randomised schedule, from 08:00 on that date. The period is given by the prefix of the record key (CTR, TRAN or INT) in the raw data.

The time interval (step) used in the models is determined by the date and time of birth. The twelve 5-week steps start at 08:00 on the Tuesdays 1 April 2025 (step 1, week 14), 6 May, 10 June, 15 July, 19 August, 23 September, 28 October and 2 December 2025, and 6 January, 10 February, 17 March and 21 April 2026 (step 12, week 17); births on the last inclusion day (26 May 2026) belong to step 12. A woman in labour at a change of period may therefore belong to a period that differs from the scheduled period of the step in which she gave birth.

Full analysis set (FAS). All women ≥ 18 years old admitted for a planned vaginal delivery in need of a partograph with a live fetus at admission at any of the participating hospitals in the period 01.04.25-26.05.26. Women delivered by caesarean section before reaching 4 cm cervical dilatation, and women who explicitly do not wish to participate in the trial, were not included. The FAS includes the women delivering during the transition periods; it is used for the analysis that includes these periods (Section 4.1.5) and to describe the trial population and flow.

Primary analysis set (PAS). The FAS excluding women delivering during their hospital's transition period (Figure 1). During the transition period staff were trained in the LCG but were asked not to act on it until the period had ended, so these women received neither condition as intended. Including them, as either condition, would bias the treatment effect towards no difference, which is anti-conservative for the non-inferiority assessment. The PAS is used for the primary analysis and for all secondary analyses unless stated otherwise.

Vaginal delivery set (VDS). The women in the PAS who delivered vaginally (spontaneous or instrumental). Used for the endpoints that exist only after a vaginal birth (episiotomy, OASIS; see Section 1.3) and for the duration component of time from 4 cm cervical dilatation to vaginal delivery (Section 5.1.1.5).

4. Analyses Supporting Primary Objective(s)

4.1 Primary Estimand

The primary estimand is the absolute difference in proportion of intrapartum caesarean sections between women with a planned vaginal delivery in need of a partograph following the LCG and the WHO partograph

4.1.1 Definition of Endpoint(s)

The endpoint is delivery by intrapartum caesarean section as recorded in the hospital's medical records. For multiple births, the woman's mode of delivery is the most operative across the babies (caesarean section, then instrumental, then spontaneous vaginal delivery), so a caesarean section for any baby counts as an intrapartum caesarean section. A caesarean section after a failed instrumental attempt is a caesarean section. If the mode of delivery of a second-born twin is not recorded, that of the first-born twin is used.

4.1.2 Main Analytical Approach

The risk difference will be estimated from a generalised linear mixed model (binomial family, logit link) with a random intercept for cluster (hospital) and fixed effects for treatment, time interval (step) and the covariates in Section 4.1.3. The marginal (population-averaged) risks are obtained by G-computation (standardisation), meaning that for every woman in the analysis set, the probability of an intrapartum caesarean section is predicted under the LCG and under the WHO partograph, with her own covariates and time interval and integrating over the distribution of the random cluster effect (Gauss–Hermite quadrature), and the predictions are averaged over all women. The absolute risk difference (primary summary measure) is the difference between the two averaged risks and the relative difference (supplementary) their ratio. Standard errors are obtained by the delta method. Because the average is taken over the time intervals of the analysis set, the estimate refers to the whole trial period.

No missing data were found for this endpoint at the blinded data review of the final data export (1 October 2026): mode of delivery was recorded for all 36,322 women, and cervical dilatation for every intrapartum caesarean section. Should the locked database nevertheless contain missing values, they will be ignored and handled by the model (complete-case analysis).

The absolute and relative differences will be presented with two-sided 95% confidence intervals, which also give the non-inferiority and superiority conclusions (Section 2.1).

4.1.3 Adjustment for covariates

The primary model is adjusted for a short, pre-specified set of strong prognostic factors for intrapartum caesarean section, to increase the precision of the standardised risk difference and to protect against changes in the population across time intervals within hospitals:

- Robson ten-group classification
- Maternal age (years)
- Maternal pre-pregnancy body mass index (kg/m^2)
- Maternal height (cm)

- Birth weight (g; first-born twin for multiple births)
- Gestational age at birth (weeks + days)

The predictability of this set was confirmed at a blinded review of the pooled data before unblinding (all study periods combined with no treatment, time interval or hospital term in the model). Robson group, maternal height, birth weight, maternal age, pre-pregnancy body mass index and gestational age were the strongest predictors of intrapartum caesarean section. Education and hypertensive disorders were borderline, and country of origin (Norway vs other) contributed little.

Continuous covariates enter as restricted cubic splines with three knots (10th, 50th and 90th percentiles); this is also the default for continuous covariates in the secondary analyses unless stated otherwise for an endpoint. The Robson classification already encodes parity, previous caesarean section, onset of labour, fetal presentation and multiple pregnancy, which are therefore not added separately; it distinguishes only term from preterm birth, so gestational age is included as a continuous covariate. Hospital-level characteristics (annual number of deliveries, intrapartum caesarean section rate) and the randomisation stratification factors are constant within a hospital and are accounted for by the random cluster effect; they are not added as covariates.

Missing covariate values are handled by the missing-indicator method: a separate “missing” category for categorical covariates, and the median plus a missing indicator for continuous covariates, so that no woman is excluded from the analysis because of a missing covariate. If fewer than 10 values of the covariate are missing, the median is imputed.

Covariate adjustment for secondary endpoints is specified for each endpoint after a blinded review of the data (Section 5). The other characteristics listed in earlier versions of this SAP (country of origin, marital status, education, smoking, snuff use, medically treated diabetes, hypertensive disorders, sex of the baby, head circumference) are presented in the baseline characteristics table but not used for adjustment, as are the annual number of deliveries of the hospital and first vaginal delivery. Derivations are given in Section 12.2.

4.1.4 Sensitivity Analyses

The sensitivity analyses keep the primary estimand and vary assumptions of the main analytical approach:

1) Unadjusted model: the model of Section 4.1.2 without the covariates of Section 4.1.3 (treatment, time interval and random cluster effect only).
2) Fixed cluster effects: hospital as a fixed effect instead of a random effect, as a check on the random-effect assumption with ten clusters.

4.1.5 Supplementary Analysis

The supplementary analyses target estimands that differ from the primary estimand in one attribute. All use the model and G-computation of Section 4.1.2.

1) Transition periods included (treatment attribute and analysis set): all women in the full analysis set are analysed, with the transition period handled as a control (WHO

partograph) period. During the transition period staff were trained in the LCG but were asked not to act on it until the period had ended.

2) Exposure time (treatment attribute): the effect of the LCG is allowed to depend on the time since the hospital started using it, with separate LCG effects for the first 5-week step after the transition period and for later steps (a two-category version of the exposure-time indicator model of Kenny et al. 2022). The rationale for this supplementary analysis is to examine whether the effect of the LCG changes as its use becomes established.

5. Analyses Supporting Secondary Objective(s)

5.1 Analyses Supporting Secondary Objectives:

5.1.1 Secondary Endpoints

C*: Continuous, possibly skewed; D: Dichotomous; Cat: Categorical; TTE: Time to event

5.1.1.1 Definition of Efficacy Endpoints for investigating the effect of LCG in women with a planned vaginal delivery

Endpoint	Measure details	Endpoint category	Handling of missing or spurious data
Spontaneous vaginal delivery	Vaginal delivery without forceps or vacuum, any presentation (including vaginal breech), as recorded in the hospital's medical records. Multiple births: spontaneous only if every baby is born spontaneously	D	No missing data observed in blind review
Instrumental vaginal delivery (forceps or vacuum)	Vaginal delivery with forceps or vacuum (combined), as recorded in the hospital's medical records. A failed instrumental attempt followed by caesarean section counts as caesarean section. Multiple births: most operative mode across the babies	D	No missing data observed in blind review
Occurrence of amniotomy during labour	Amniotomy, as recorded in the hospital's medical records, whatever the cervical dilatation at which it was performed	D	Blind review: 1 woman with missing amniotomy status imputed as no
Use of oxytocin during labour	Oxytocin infusion (including the second stage), as recorded in the hospital's medical records, whatever the cervical dilatation at which it was started	D	Blind review: 1 woman with missing oxytocin status imputed as no

Endpoint	**Measure details**	**Endpoint category**	**Handling of missing or spurious data**
Use of epidural analgesia during labour	Epidural analgesia, as recorded in the hospital's medical records, whatever the cervical dilatation at which it was started	D	Blind review: 16 women with missing epidural status imputed as no
Intermittent fetal monitoring	Exclusive intermittent fetal monitoring in labour: intermittent monitoring recorded and neither continuous external CTG nor STAN/continuous internal CTG. Women converted to continuous monitoring are not events. Continuous external CTG is assumed to include the admission CTG, so exclusive intermittent monitoring may be understated	D	Blind review: 107 'none' and 2 with nothing ticked, counted as no event; a ticked intermittent box overrides 'none'
Continuous fetal monitoring	Any continuous fetal monitoring in labour: continuous external CTG or STAN/continuous internal CTG recorded (with or without intermittent monitoring). Continuous external CTG is assumed to include the admission CTG	D	Blind review: 107 'none' and 2 with nothing ticked, counted as no event
Episiotomy	Use of episiotomy in labour.	D	Blind review: 8 women with missing status imputed as no Vaginal delivery set (VDS)
Occurrence of OASIS	Obstetric anal sphincter injury: perineal tear of grade 3 (3a–3c) or 4, as recorded in the hospital's medical records	D	Blind review: 362 (1.2%) with no tear status or a tear without

Endpoint	Measure details	Endpoint category	Handling of missing or spurious data
			grade; complete case (CC) Vaginal delivery set (VDS)
Occurrence of post-partum haemorrhage > 1000 ml	Estimated blood loss > 1000 ml within 2 hours after birth, as recorded in the hospital's medical records. Blood loss of exactly 1000 ml (a common rounded value) is not an event	D	Blind review: 181 women (0.5%) with no blood loss recorded; complete case (CC)
Time from 4 cm cervical dilation to vaginal delivery	Time from 4 cm cervical dilation until vaginal delivery (birth of the first baby), restricted to the first 36 hours: min(T, 36 h) after a vaginal delivery, 36 h after a caesarean section. Time zero is the vaginal examination at which 4 cm was found; if no examination found 4 cm, the time of reaching 4 cm is multiply imputed (see text)	TTE	Caesarean section: 36 h (composite, see text). Missing 4 cm time: multiple imputation (MAR)

C*: Continuous, possibly skewed; D: Dichotomous; Cat: Categorical; TTE: Time to event

5.1.1.2 Definition of efficacy Endpoints for investigating the effect of LCG on the neonate

Endpoint	Measure details	Endpoint category	Handling of missing data
Apgar score < 7 at 5 minutes after birth	The Apgar score < 7 evaluated at 5 minutes after birth in any baby (worst baby per woman for multiple births, among the babies with a score) as recorded in the hospital's medical records; stillborn babies as recorded	D	Blind review: 4 first-born and 3 second-born twins without a score; complete case (CC)

Endpoint	Measure details	Endpoint category	Handling of missing data
Admittance to neonatal intensive care unit within 2 hours after birth	Any admittance of any duration of any baby to the neonatal intensive care unit within 2 hours after birth as recorded in the hospital's medical records (worst baby per woman; a stillborn baby is not admitted)	D	No missing data observed in blind review
Arterial umbilical cord pH < 7.00	Arterial umbilical cord pH < 7.00 in any baby (worst baby per woman for multiple births, among the measured babies)	D	Blind review: arterial pH not recorded in 27.6%; multiple imputation (MAR), see text

C*: Continuous, possibly skewed; D: Dichotomous; Cat: Categorical; TTE: Time to event

5.1.1.3 Definition of Descriptive or Exploratory Endpoints for investigating the effect of LCG in women

Endpoint	Measure details	Endpoint category	Handling of missing data
Cervix dilation at time of caesarean section	Cervix dilation in cm at time of caesarean section	C*	Descriptive; not recorded shown as missing
Level of the presenting part at caesarean section	Level of the presenting part at caesarean section Above the ischial spines, at the level of the ischial spines, below the ischial spines	Cat	Descriptive; not recorded shown as missing
Fetal presentation at caesarean section	Fetal presentation at caesarean section Breech, occiput (vertex), sinciput, brow or face	Cat	Descriptive; not recorded shown as missing
Fetal position at caesarean section	Fetal position at caesarean section	Cat	Descriptive; not recorded shown as missing

Endpoint	Measure details	Endpoint category	Handling of missing data
	Occiput anterior, occiput posterior or transverse		
Main reason for caesarean section	Main reason for caesarean section Fetal distress, prolonged first or second stage of labour, failed induction, failed instrumental vaginal delivery, breech, maternal consideration, others	Cat	Descriptive; not recorded shown as missing
Type of continuous fetal monitoring	• Continuous external CTG • STAN/continuous internal CTG	Cat	Descriptive; not recorded shown as missing
Occurrence of post-partum haemorrhage	Any occurrence of post-partum haemorrhage >500 ml, >1000 ml, >1500 ml	Cat	Descriptive; not recorded shown as missing
Occurrence of prolonged labour	Occurrence of prolonged labour defined as: **WHO partograph:** First stage: Crossing of the action line Second stage: 2. stage lasting from 2-4 hours, depending on the labour wards guidelines **LCG:** First stage: Time intervals exceeded Second stage: Time intervals exceeded	D	Descriptive; not recorded shown as missing

Endpoint	Measure details	Endpoint category	Handling of missing data
Shared decision making	Shared decision making for operative delivery, documented by the midwife: whether an operative delivery was performed, and whether the woman was informed and in agreement. Items, Norwegian wording and response options: Section 12.2. Recorded on the LCG form only; presented for the LCG period. The follow-up questions are presented among the women with an operative delivery recorded on this item.	D+Cat	Descriptive; not recorded shown as missing

5.1.1.4 Main Analytical Approach for Dichotomous Efficacy Endpoints

Dichotomous endpoints are analysed as the primary endpoint (Section 4.1.2): a generalised linear mixed model (binomial family, logit link) with a random intercept for cluster and fixed effects for treatment, time interval and the covariates specified for the endpoint, with the absolute and relative differences obtained by G-computation integrating over the random cluster effect and standard errors by the delta method. Covariates are specified per endpoint after the blinded data review (Section 4.1.3). Endpoints restricted to vaginal births (episiotomy, OASIS) are analysed in the vaginal delivery set, all others in the primary analysis set (Section 3). Missing data are handled as given in the "Handling of missing data" column of the endpoint tables.

The differences are presented with two-sided 95% confidence intervals, not adjusted for multiplicity (Section 2.2). No hypothesis tests are performed and no non-inferiority margin applies to secondary endpoints.

Covariate sets by endpoint:

- Spontaneous vaginal delivery and instrumental vaginal delivery: as for the primary endpoint (Section 4.1.3). Mode of delivery is classified as spontaneous vaginal, instrumental vaginal or caesarean section, so the three differences sum to zero. The difference in caesarean sections is the primary analysis (Section 4.1.2).
- Occurrence of amniotomy during labour: as for the primary endpoint.
- Use of oxytocin during labour: as for the primary endpoint.
- Use of epidural analgesia during labour: as for the primary endpoint.
- Intermittent fetal monitoring: as for the primary endpoint.

- Continuous fetal monitoring: as for the primary endpoint.
- Episiotomy: as for the primary endpoint.
- Occurrence of OASIS: as for the primary endpoint.
- Occurrence of postpartum haemorrhage > 1000 ml: as for the primary endpoint.
- Arterial umbilical cord pH < 7.00: as for the primary endpoint.
- Apgar score < 7 at 5 minutes: as for the primary endpoint.
- Admittance to neonatal intensive care unit within 2 hours: as for the primary endpoint.

Arterial umbilical cord pH was not recorded for 27.6% of the first-born babies at the blinded review (venous pH only for 5,328 women, no cord blood gas for 4,697). Missing values are assumed to be missing at random: whether the arterial pH was recorded may depend on the hospital, the time interval, the mode of delivery and the condition of the baby as reflected in the auxiliary variables (venous pH, base excess, Apgar scores, admission to the neonatal intensive care unit), but not on the unrecorded value itself given these. All these variables are in the imputation model. When no baby of a woman has a measured arterial pH, the arterial pH of the first-born baby is imputed by multiple imputation under a missing-at-random assumption (fully conditional specification with predictive mean matching, 30 imputed datasets) and dichotomised at 7.00. The imputation model contains all variables of the analysis model (treatment, time interval, hospital as a fixed effect and the covariates) and the auxiliary variables venous cord pH, arterial and venous base excess or deficit, Apgar scores at 1 and 5 minutes, admission to the neonatal intensive care unit within 2 hours and mode of delivery. For multiple births with at least one measured baby, the endpoint is taken from the measured babies. Estimates are combined by Rubin's rules. Arterial and venous pH values outside 6.5–7.55, and base excess or deficit outside −25 to 25 mmol/l, are considered not plausible and are set to missing before imputation.

5.1.1.5 Main Analytical Approach for Time to Event Efficacy Endpoints

Time from 4 cm cervical dilatation to vaginal delivery is analysed as the time without vaginal delivery during the first 36 hours after 4 cm, in the primary analysis set (Section 1.3). For each woman, let T be the time from 4 cm to delivery (the birth of the first baby for multiple births). The analysis variable is Y = min(T, 36 h) after a vaginal delivery (spontaneous or instrumental) and Y = 36 h after a caesarean section, whenever it occurs. Mode of delivery is the most operative across the babies (Section 4.1.1), so a caesarean section for a second twin counts as a caesarean section.

The mean of Y is the restricted mean time without vaginal delivery: 36 hours minus the area under the cumulative incidence curve of vaginal delivery up to 36 hours (Andersen 2013). The summary measure is the difference in mean Y between the LCG and the WHO partograph, in hours. Caesarean section is handled by a composite strategy (Section 1.3), so the estimand is the total effect of the LCG on reaching a vaginal delivery over time, including any effect through caesarean section (Young et al. 2020).

The horizon of 36 hours is longer than virtually all times to vaginal delivery (at the blinded review, among the women with an examination at 4 cm, 1.5% of the times from 4 cm to delivery exceeded 24 hours and 0.2% exceeded 36 hours), so that a vaginal delivery is almost never

truncated and a caesarean section counts as at least as unfavourable as the longest labours ending in a vaginal delivery. Beyond the vaginal deliveries, the horizon sets the weight given to a caesarean section: with a mean time from 4 cm to vaginal delivery of about 7.5 hours, a difference of one percentage point in caesarean sections changes the summary measure by about 0.3 hours. Horizons of 24 and 48 hours are supplementary analyses (Section 5.1.1.7).

There is no censoring, every woman is followed until delivery. Time zero is the vaginal examination at 4 cm cervical dilatation. For women who deliver vaginally without an examination at 4 cm, the time of reaching 4 cm is multiply imputed from a labour-curve model fitted to the vaginal deliveries. Following the reverse approach of Zhang et al. (2002, 2010), who constructed average labour curves with time measured backwards from full dilatation, because women are admitted at different dilatations but all reach the same end point, time is measured backwards from delivery, which is recorded for every woman: the logarithm of the time from each examination to delivery is modelled as a natural cubic spline of the dilatation found (3 degrees of freedom), with a random intercept and slope per woman, fixed effects for treatment, the interaction of treatment with dilatation, time interval, hospital and the covariates of the primary endpoint, and a single residual variance. For each woman, the time of reaching 4 cm is drawn from her predictive distribution given her own examinations, truncated to lie between her last examination below 4 cm (if any) and her first examination at 4 cm or more, with the model parameters drawn anew for each of 30 imputed datasets. A vaginal delivery more than 36 hours after the observed or imputed time of reaching 4 cm contributes 36 hours.

Because Y is known for every woman once time zero is imputed, the summary measure is estimated as a difference in means, without a model for the distribution of the times. Y is analysed with a linear mixed model with a random intercept for cluster and fixed effects for treatment, time interval and the covariates of the primary endpoint (Section 4.1.3). With the identity link, the treatment coefficient is the marginal difference in mean Y, and no G-computation is needed. Because no woman is censored, the mean of Y equals 36 hours minus the area under the empirical cumulative incidence curve of vaginal delivery, and the regression of Y on treatment and covariates is the pseudo-observation approach to the restricted mean (Andersen, Hansen and Klein 2004), with the pseudo-observations equal to Y. Y is not normally distributed (it has a point mass at 36 hours from the caesarean sections); the estimate does not rely on normality, and with more than 30,000 women the confidence interval relies on the central limit theorem. The model is fitted to each of the 30 imputed datasets, and the estimates and their model-based variances are combined by Rubin's rules. The difference is presented with a two-sided 95% confidence interval; no hypothesis test is performed (Section 5.1.1.4).

To show whether a difference comes from caesarean sections or from the duration of labour, two components are presented: the difference in the probability of vaginal delivery, which is minus the difference in caesarean sections of the primary analysis (Section 4.1.2), and the difference in the mean of min(T, 36 h) among women who deliver vaginally, from the same model in the vaginal delivery set. The latter conditions on vaginal delivery and is not interpreted as a treatment effect on its own (Section 1.3).

As supportive analyses, the cumulative incidence of vaginal delivery by time since 4 cm is shown for each treatment (averaged over the imputed datasets; descriptive, not adjusted for time interval), and the differences in the proportion of women delivered vaginally within 6 and within

12 hours of 4 cm are estimated as for the dichotomous endpoints (Section 5.1.1.4) in each imputed dataset and combined by Rubin's rules.

5.1.1.6 Sensitivity Analyses

For time from 4 cm cervical dilatation to vaginal delivery, two sensitivity analyses keep the estimand and vary the model of Section 5.1.1.5:

1) Unadjusted model: treatment, time interval and random cluster effect only.
2) Fixed cluster effects with robust standard errors: hospital as a fixed effect instead of a random effect, with heteroscedasticity-consistent (HC3) standard errors, which do not assume a constant residual variance. The variance of Y depends strongly on the risk of caesarean section because of the point mass at 36 hours.

There will be no sensitivity analyses for the other secondary endpoints.

5.1.1.7 Supplementary Analyses

For the endpoints analysed in the vaginal delivery set (episiotomy, OASIS), a supplementary analysis uses a composite strategy in the primary analysis set, counting a caesarean section as no event (no episiotomy, no OASIS) (Section 1.3).

For time from 4 cm cervical dilatation to vaginal delivery, supplementary analyses restrict the time without vaginal delivery to the first 24 and to the first 48 hours after 4 cm instead of 36 hours, with the analysis of Section 5.1.1.5. The horizon sets the weight given to a caesarean section, so each horizon defines a different estimand.

6. Analyses Supporting Exploratory Objectives

Hypothesis generating exploratory analyses will be performed on exploratory endpoints. These analyses will not be detailed in this analysis plan and will be regarded as post-hoc analyses.

Descriptive endpoint, especially categorical endpoints may be presented descriptively in the main statistical report (Section 5.1.1.3).

The pre-specified subgroup analysis is described in Section 8.2.

7. Safety Analyses

There are no separate safety analyses. The maternal and neonatal safety outcomes — postpartum haemorrhage > 1000 ml, Apgar score < 7 at 5 minutes, arterial umbilical cord pH < 7.00 and admission to the neonatal intensive care unit — are secondary endpoints and are analysed in Section 5.

7.1 Extent of Exposure

Extent of exposure is not applicable: the intervention is a labour-monitoring tool used at the unit level, and treatment is defined by the eCRF (Section 3). The use of the LCG in practice is described descriptively through shared decision making for operative delivery (Section 5.1.1.3); health care providers' experience with the LCG is reported in a separate publication (Section 12.1).

7.2 Adverse Events

Adverse events are not collected systematically and are not analysed. Serious and unexpected adverse events are reported by the principal investigator to the ethics committee (protocol section 9.1). Serious maternal and neonatal outcomes are covered by the secondary endpoints (Section 5).

7.3 Additional Safety Assessments

Not applicable.

8. Other Analyses

8.1 Other Variables and/or Parameters

Not applicable

8.2 Subgroup Analyses

One exploratory subgroup analysis is prespecified, for the primary endpoint by Robson group (Ten Group Classification System). The model of Section 4.1.2 is extended with a treatment × Robson group interaction, and the absolute risk difference is estimated within each group by G-computation over the women of that group, with two-sided 95% confidence intervals (delta method). Heterogeneity is assessed by a likelihood-ratio test of the interaction. Robson groups with fewer than 50 intrapartum caesarean sections in the analysis set (both periods together, known before unblinding) are combined into one group. The results are exploratory, not adjusted for multiplicity, and no non-inferiority conclusion is drawn within subgroups. Hospital-level characteristics (hospital, annual number of deliveries, intrapartum caesarean section rate) are not used for subgroup analyses: with ten clusters, each subgroup would contain only a few hospitals.

If there are indications of intervention-modifying baseline variables, further subgroup analyses may be included as exploratory analyses.

9. Interim Analyses

There will be no interim analyses.

10. Changes to Protocol-planned Analyses

The following changes to the analyses planned in protocol version 1.06 were decided before database lock and without access to unblinded data.

Primary hypothesis. The protocol describes a non-inferiority approach in its design section but calculates the sample size for equivalence. This SAP specifies non-inferiority, with a margin of 1.5 percentage points for the absolute risk difference (Section 2.1). This does not reduce the power of the planned sample size (Section 11).

Summary measure. The protocol presents the difference between groups as a risk ratio. The non-inferiority margin is defined on the absolute scale, so the absolute risk difference is the primary summary measure and the relative risk is supplementary; both are marginal (population-averaged) estimates obtained by G-computation (Section 4.1.2).

Tests for secondary endpoints. The protocol plans a significance test accounting for the cluster structure for the dichotomous efficacy variables. Secondary endpoints are reported with 95% confidence intervals without hypothesis tests and without adjustment for multiplicity (Section 2.2).

Covariate adjustment. The protocol's model contains the random cluster effect, treatment and time interval. The prespecified prognostic covariates of Section 4.1.3 are added to increase precision and to protect against changes in case mix across time intervals; the unadjusted model is a sensitivity analysis.

Estimands and analysis populations. The protocol specifies an intention-to-treat analysis. This SAP specifies the estimands according to ICH E9(R1) (Section 1.1): a treatment policy strategy for non-adherence; treatment defined by the eCRF, which changed at the randomised crossover dates (Section 3); the transition periods excluded from the primary analysis set and included in a supplementary analysis; all caesarean sections from 4 cm counted for the primary endpoint in both periods; a principal stratum strategy for episiotomy and OASIS; and a composite strategy for time from 4 cm cervical dilatation to vaginal delivery.

Subgroup analyses. The protocol plans subgroup analyses of specific elements of the LCG (start of active labour at 5 cm, labour progression assessed by time intervals, and shared decision making). These elements are features of the LCG period after randomisation, not baseline characteristics, so they do not define subgroups in which the randomised comparison is preserved, and they are not analysed as subgroups. The different start of active labour (4 cm with the WHO partograph, 5 cm with the LCG) is handled by counting all caesarean sections from 4 cm, and shared decision making for operative delivery is described (Section 5.1.1.3). One exploratory subgroup analysis by Robson group is added (Section 8.2).

As-treated and per-protocol analyses. The protocol plans as-treated or per-protocol analyses with inverse probability weighting or g-estimation in case of substantial lack of adherence due to missing data. These will not be performed, there is no individual measure of adherence, and missing data on the primary endpoint are negligible (Section 4.1.2).

Endpoints. The protocol's labour duration is specified as time from 4 cm cervical dilatation to vaginal delivery, analysed as the restricted mean time without vaginal delivery (Section 5.1.1.5). The protocol's perineal tears are analysed here as the occurrence of obstetric anal sphincter injury (OASIS); the degree of perineal tears is analysed in a later statistical analysis plan. Labour

progression, childbirth experience, perineal support, medical pain relief and other secondary endpoints of the protocol are analysed in later statistical analysis plans (Section 12.1); this is a matter of scope, not a change of analysis.

11. Sample Size Determination

From the protocol: The power calculation is based on the average rate of intrapartum caesarean section of 10.5 % in the NORBIRTH network. If the 95% confidence interval of the difference in the proportions of caesarean section between the LCG and the WHO partograph lies completely within -delta to +delta margins of equivalence, then the two treatments are considered equivalent. With a power of 80%, a significance level (alpha) of 5% and a delta margin of equivalence of ± 1.5 percentage points, the analysis need to include 23 650 women considered the cluster effect and an estimated intraclass correlation coefficient (ICC) of approximately 0.0001 as described by Henrichs et al. (19). Within the NORBIRTH network approximately 28.950 women use a partograph each year. To secure a sufficient number of individuals in the analyses, the inclusion period is estimated to last for one year and two months (60 weeks). Each hospital will cross from WHO partograph to the LCG at equal time intervals until all hospitals have crossed, which corresponds to five weeks per step taken into account a normal variation of annual deliveries and potential opt outs.

The primary hypothesis is non-inferiority (Section 2.1). The sample size above was calculated for equivalence with the same margin, i.e. for two one-sided tests. The non-inferiority test is one of these, so its power is at least that of the equivalence calculation, and the trial is, if anything, over-powered for the non-inferiority claim. The number of women included also exceeds the planned 23 650.

12. Supporting Documentation

Table and figure layouts are given by the mock-up statistical report, produced with the analysis code on an anonymised copy of the data.

12.1 Other objectives and endpoints specified in the protocol, detailed elsewhere

Objectives	**Endpoints**
Secondary	
To investigate the childbirth experience in women with a planned vaginal delivery	Total and mean scores of the four domains (Own capacity, Professional support, Perceived safety and Participation) of the Childbirth Experience Questionnaire (CEQ2)
To investigate the effect of LCG in women with a planned vaginal delivery	Total dosage of oxytocin
	Maximum dosage of oxytocin
	Total duration of oxytocin
	Initiation of oxytocin
	Use of medical pain relief in labour
	Total duration of epidural analgesia
	Use of non-medical pain relief in labour
	Occurrence of continuous support in labour
	Occurrence of perineal support
	Occurrence of perineal tears
	Degree of perineal tears
	Total estimated blood loss
	Occurrence of postpartum blood transfusion

Objectives	**Endpoints**
	Cervix dilatation at induction of labour
	Labour progression: time from one centimetre to the next, starting at 5 cm, until 10 cm
	Time related aspects of labour duration
	Breastfeeding at discharge
	Reason for admission
	Supportive care, labour companionship
	Type of perineal support
To investigate the effect of LCG on the neonate	Apgar scores at 1, 5 and 10 minutes
	Neonatal morbidity
	Occurrence of neonatal metabolic acidosis (arterial umbilical cord pH < 7 and BE ≤ -12 or BD ≥ 12)
	Arterial umbilical cord pH
	Occurrence of undisturbed skin-to-skin contact for at least one hour after birth
	Breastfeeding within two hours after birth
	The use of formula milk
Exploratory	
To describe and explore the effect of LCG in women with a planned vaginal delivery	Fetal presentation at time of delivery
	Fetal position at time of delivery
	Initiation of epidural analgesia

Objectives	Endpoints
	Birthing position
	Women's childbirth experiences through qualitative interviews
To describe and explore the effect of LCG on the neonate	Timing of cord clamping
To describe and explore the use of LCG from the health care providers perspective	Health care providers experiences with using the LCG in clinical practice through a survey and through qualitative interviews

12.2 Data derivations

Derivation of covariates and baseline characteristics from the trial database (variable names as in the data export). For multiple births, neonatal characteristics refer to the first-born twin.

Characteristic	Source variable(s)	Derivation
Annual number of deliveries	metadata/hospitals.csv (births per year at the hospital)	1000–1999, 2000–3999, ≥ 4000
Maternal age	alder	Age at the time of delivery in completed years (from date of birth), as recorded
Country of origin	landbakgrunn	Grouped by continent (country-to-continent lookup)
Country of origin Norway	landbakgrunn	Norway vs other
Married or cohabiting	sivilstatus	Married (gift) or cohabiting (samboer) vs other
Higher education	utdanning	University or college (hoyere_utdanning) vs lower
Height	hoyde	cm; adjustment covariate (Section 4.1.3), not presented in the baseline characteristics table

Characteristic	**Source variable(s)**	**Derivation**
Pre-pregnancy weight	pregravid_vekt	kg
Pre-pregnancy BMI	hoyde, pregravid_vekt	weight (kg) / height $(m)^2$
Smoking	royk_for_eller_i_sv_s	Smoking before or during pregnancy
Snuff use	snus_for_eller_i_sv_s	Snuff use before or during pregnancy
Medically treated diabetes	hvilke_kjente_pregravide_sykdommer, hvilke_svangerskapsrelaterte_sykdommer	Type 1 or type 2 diabetes, or gestational diabetes treated with metformin or insulin
Hypertensive disorders	as above	Chronic hypertension, gestational hypertension or pre-eclampsia
Robson group	robson_gruppe	As recorded (ten-group classification)
Parity	paritet	0, 1, ≥ 2
Previous caesarean section	tidligere_sectio	As recorded
First vaginal delivery	paritet, tidligere_sectio	Nulliparous, or para 1 with a previous caesarean section (para ≥ 2 with a previous caesarean section counted as no: not derivable)
Spontaneous onset of labour	induksjon	No induction (induksjon = 0)
Induction of labour	induksjon	induksjon = 1
Gestational age	gestasjonsalder_ved_fodsel_hele_uker, ..._pluss_dager	Weeks + days at birth
Sex of the baby	barnets_kjonn	As recorded

Characteristic	Source variable(s)	Derivation
Birth weight	fodselsvekt_gram	g
Birth weight category	fodselsvekt_gram	< 3000, 3000–3499, 3500–3999, 4000–4499, ≥ 4500 g
Head circumference	neonatal_hodeomkrets	cm

The English labels of all coded values of the export are kept in a validated codelist file with the analysis code.

Shared decision making is recorded on the LCG form only. The items for operative delivery, their wording on the form and the condition under which each follow-up question is asked are given below.

Table 3: Shared decision-making items for operative delivery (LCG form). † Norwegian wording reconstructed from the form's pattern for items not in the form version of 9 March 2025; to be verified.

Item	Wording on the form	Responses	Asked if yes to
Operative delivery: was the intervention performed?	Operativ forløsning †	yes/no	
Operative delivery: was the woman informed?	Hvis operativ forløsning - ble kvinnen informert om tiltaket	yes/no	Operative delivery: was the intervention performed?
Operative delivery: did she agree?	Hvis operativ forløsning - var hun enig i tiltaket	yes/no/not asked/no answer	Operative delivery: was the intervention performed?

13. References

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